Program: **Diploma in Artificial Intelligence & Machine Learning**	
Course Code: **6342C**	Course Title: **Big Data Concepts**
Semester: **6**	Credits: **4**
Course Category: **Open Elective**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To familiarize with Big Data and its sources
- To understand the basics of data preprocessing and data visualization
- To familiarize Big Data tools and techniques.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Mathematics		Mathematics I & II	1,2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize the concepts of Big Data	12	Understanding
CO2	Outline the basic concepts in Big Data Analytics	15	Understanding
CO3	Implementation of Exploratory Data Analysis using Python	15	Applying
CO4	Demonstrate Map Reduce programming in Hadoop Environment	16	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Descriptions	Duration (Hours)	Cognitive Level
CO1	**Familiarize the concept of Big Data**		
M1.01	Familiarize the concepts of Big Data.	3	Understanding
M1.02	Describe Big data Analytics and its applications	3	Understanding
M1.03	Overview of techniques used in Data Science	3	Understanding
M1.04	Compare Big Data and Data Warehouse	3	Understanding
Contents: **Introduction to Bigdata :** Data –Big data – concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. **Data Science/ Big Data Analytics :** Definition, Requirement of high performance analytics, benefits, applications and challenges. **Techniques used in Data Science -** classification, regression and clustering. **Data Warehouse:** Definition, Coexistence of Big Data and Data Warehouse.			
CO2	**Outline the basic concepts in Big Data Analytics**		
M2.01	Summarize KDD Process.	3	Understanding
M2.02	Explain data mining techniques	5	Understanding
M2.03	Describe data preprocessing methods	4	Understanding
M2.04	Outline data visualization.	3	Understanding
	Series Test – I	1	
Contents:			

Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. **Data Mining-** Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and Genetic Algorithm. Knowledge Representation Methods **Data Preprocessing** - Need of data preprocessing, data quality, data preprocessing methods-cleaning, integration, transformation, reduction, discretization, normalization. **Data Visualization:** Definition, Need for Visualization, Advantages.	
CO3	**Implement of Exploratory Data Analysis using Python**

M3.01	Familiarize the EDA process.	3	Understanding
M3.02	Illustrate step-by-step Exploratory Data Analysis (EDA) using Python.	8	Applying
M3.03	Summarize the concepts of NoSQL	2	Understanding
M3.04	Outline the concepts of NewSQL	2	Understanding

Contents:

EDA process: Definition, importance, types-univariate, bivariate and multivariate.

Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks)

Import python libraries - numpy, pandas, matplotlib, seaborn.

Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation.

Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps).

NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data.

NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL

CO4	**Demonstrate Map Reduce programming in Hadoop Environment**		
M4.01	Explain the fundamentals of Hadoop	2	Understanding
M4.02	Concepts of HDFS	2	Understanding
M4.03	List the components of Hadoop Ecosystem	2	Understanding
M4.04	Illustrate the concept of HDFS and MapReduce	4	Applying
M4.05	Use the different function in MapReduce	3	Applying
M4.06	Outline the basic concepts of Hive and Pig	3	Understanding
	Series Test -2	1	

Contents:

Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions

Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing

Data with Hadoop

Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting.

Basic concepts of Hive and Pig

Text / Reference

T/R	Book Title/Author
T1	T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015
T2	Jiawei Han,Champaign Micheline Kamber ‚Jian Pei Simon Fraser ‘”Data Mining Concepts and Techniques Third Edition”
R1	Frank J Ohlhorst, ―Big Data and Analytics: Turning Big Data into Big Money, Wiley and SAS Business Series, 2012.
R2	Tom White, ― Hadoop: The Definitive Guide Third Edition, O‘reily Media, 2012
R3	EMC Education Services, Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012.

Online Resources

Sl.No	Website Link
1	https://hadoop.apache.org/
2	https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/
4	https://hive.apache.org/
5	https://www.tutorialspoint.com/hadoop/index.htm
6	https://pig.apache.org/